HUJI · 67731 · OPTIMIZATION · 2026-27

Week 02

Calculus and Optimization

Jacobian · gradient · chain rule · Hessian · Taylor · local vs. global minima · convexity in one dimension

What this PDF covers. Lecture 2 only (§2.1 of Wiesel's notes); the recitation is not included. The last part has **16 practice problems with full solutions**, marked by difficulty ●○○ (easy) to ●●● (hard), and a cheat sheet. **Reading in Boyd & Vandenberghe** [BV]: App. A.4.1–A.4.4, §1.1, §4.1.1, §2.1.4, §3.1.1, §3.1.3–3.1.5 (functions on $\mathbb{R}$), §4.2.2.

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PART A · LECTURE 2

Calculus and Optimization

Optimization means: find the input x that makes a function as small as possible. Calculus is the tool that tells us, without trying every x, which way is "downhill" (the gradient) and how the ground curves (the Hessian). This lecture builds the tools (§1), sets up what an optimization problem is (§2), and then shows why *convexity* is the property that makes optimization tractable (§3).

SECTION 01

Multivariate calculus

> 1.1 Vocabulary

DEF · A FUNCTION $f : \mathbb{R}^n \to \mathbb{R}^m$

A rule that takes n numbers in and gives m numbers out: $f(x) = (f_1(x), \dots, f_m(x))$, where each $f_i : \mathbb{R}^n \to \mathbb{R}$ is one output. Example: $f(x_1, x_2) = (x_1^2 x_2,\ x_1 + x_2^2)$ has $n = 2$ inputs and $m = 2$ outputs; $f(1, 2) = (2, 5)$.

DEF · PARTIAL DERIVATIVE

$\dfrac{\partial f_i}{\partial x_j}(x)$ is the ordinary derivative of output i with respect to input j, *holding all other inputs fixed*. Example: for $f_1 = x_1^2 x_2$, treat x_2 as a constant: $\partial f_1/\partial x_1 = 2x_1 x_2$; treat x_1 as a constant: $\partial f_1/\partial x_2 = x_1^2$.

> 1.2 The Jacobian

DEF · JACOBIAN

The Jacobian of $f : \mathbb{R}^n \to \mathbb{R}^m$ at x is the $m \times n$ matrix of all partial derivatives:

$$Df(x) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix}, \qquad [Df(x)]_{ij} = \frac{\partial f_i(x)}{\partial x_j}.$$

Row i = output i, column j = input j. So it has as many rows as outputs and as many columns as inputs.

EXAMPLE · A JACOBIAN WITH NUMBERS

$f(x_1, x_2) = (x_1^2 x_2,\ x_1 + x_2^2)$:

$$Df(x) = \begin{pmatrix} 2x_1x_2 & x_1^2 \\ 1 & 2x_2 \end{pmatrix}, \qquad Df(1,2) = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}.$$

EXAMPLE · EXAMPLE 2.1: A LINEAR MAP, $f(x) = Ax$

Output i is $f_i(x) = \sum_k A_{ik}x_k$. Differentiating with respect to x_j kills every term except $k = j$:

$$[Df(x)]_{ij} = \frac{\partial}{\partial x_j} \sum_k A_{ik}x_k = A_{ij} \quad \Longrightarrow \quad Df(x) = A.$$

Exactly like the scalar rule $(ax)' = a$.

What the Jacobian is for: the first-order approximation. Near a point x, a smooth function looks like a linear map:

FIRST-ORDER APPROXIMATION

$$f(z) \approx f(x) + Df(x)\,(z - x).$$

Sizes: $f(z)$ is $m \times 1$, $Df(x)$ is $m \times n$, $(z - x)$ is $n \times 1$. If the sizes do not match, you have a transpose wrong.

For $f(x) = Ax$ this is *exact*, $Az = Ax + A(z - x)$, which is a handy way to remember the sizes.

EXAMPLE · HOW GOOD IS THE APPROXIMATION?

Same f as above, at $x = (1, 2)$, step to $z = (1.1,\ 1.9)$, so $z - x = (0.1, -0.1)$:

$$f(z) \approx \begin{pmatrix} 2 \\ 5 \end{pmatrix} + \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 0.1 \\ -0.1 \end{pmatrix} = \begin{pmatrix} 2.3 \\ 4.7 \end{pmatrix}.$$

Exact: $f(1.1, 1.9) = (1.21 \cdot 1.9,\ 1.1 + 3.61) = (2.299,\ 4.71)$. The error is about 0.01, i.e. of the order of $(\text{step})^2$.

> 1.3 The gradient

When there is *one* output ($m = 1$, the case of an objective function), the Jacobian is a single row $1 \times n$. We stand it upright:

DEF · GRADIENT

For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is the transposed Jacobian, always a column vector:

$$\nabla f(x) = [Df(x)]^T = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)^T, \qquad f(z) \approx f(x) + \nabla f(x)^T (z - x).$$

WHAT THE GRADIENT TELLS YOU

$\nabla f(x)^T(z-x)$ is the predicted change of f when you move by $z-x$. It is largest when the step points along $\nabla f(x)$ (Cauchy–Schwarz, Week 01), so the gradient points in the direction of steepest ascent, and $-\nabla f$ is steepest descent. This is the idea behind gradient descent (Chapter 4).

EXAMPLE · EXAMPLE 2.2: INNER PRODUCT, $f(x)=a^Tx$

$f(x)=\sum_i a_ix_i$, so $\partial f/\partial x_j = a_j$: $\quad Df(x)=a^T$ (a row), $\quad \nabla f(x)=a$ (a column).

> 1.4 The gradient of a quadratic form

$f(x)=x^TAx$ (with A an $n\times n$ matrix) appears everywhere: energy, variance $w^T\Sigma w$, least squares. The lecture finds its gradient *without* writing out partial derivatives, by using the first-order approximation as a definition.

Step by step.

1. Move from x a small amount δ in a direction z. The approximation says $f(x+\delta z)\approx f(x)+\delta\,\nabla f(x)^Tz$. So whatever multiplies δ is $\nabla f(x)^Tz$.
2. Expand exactly: $(x+\delta z)^TA(x+\delta z)=x^TAx+\delta\,(z^TAx+x^TAz)+\delta^2\,z^TAz$.
3. Ignore the δ^2 term (second order). Matching the δ terms: $\nabla f(x)^Tz=z^TAx+x^TAz$.
4. Make both terms look like "something^Tz". A scalar equals its transpose, so $z^TAx=(z^TAx)^T=x^TA^Tz$. Hence
$$\nabla f(x)^Tz=x^TA^Tz+x^TAz=\big[(A+A^T)x\big]^Tz \quad \text{for every } z.$$
5. The only vector that gives this inner product with every z is $\nabla f(x)=(A+A^T)x$.

EXAMPLE 2.3: QUADRATIC FORM

$$\nabla(x^TAx)=(A+A^T)x, \qquad \text{and if } A \text{ is symmetric:} \quad \nabla(x^TAx)=2Ax.$$

Compare the scalar case $(ax^2)'=2ax$.

EXAMPLE · CHECKING WITH NUMBERS

$A=\begin{pmatrix}2&1\\1&2\end{pmatrix}$ (symmetric), $x=(1,0)$. Then $f(x)=2$, $\nabla f(x)=2Ax=(4,2)$. Step $\delta=0.1$ in direction $z=(0,1)$: prediction $2+0.1\cdot(4,2)\cdot(0,1)=2.2$. Exact: $f(1,0.1)=2\cdot1+2\cdot1\cdot0.1+2\cdot0.01=2.22$. The difference 0.02 is the dropped δ^2z^TAz term. ✓

PITFALL · $2Ax$ ONLY FOR SYMMETRIC A

For $A=\begin{pmatrix}1&2\\0&3\end{pmatrix}$ and $x=(1,1)$: $f=x_1^2+2x_1x_2+3x_2^2$, whose partial derivatives give $\nabla f=(2x_1+2x_2,\ 2x_1+6x_2)=(4,8)$. The formula $(A+A^T)x$ gives $(4,8)$ ✓; the shortcut $2Ax$ gives $(6,6)$ ×. (Any quadratic form can be rewritten with the symmetric matrix $\frac{1}{2}(A+A^T)$,

which is why we usually assume symmetry.)

> 1.5 The chain rule

CHAIN RULE

For $f : \mathbb{R}^n \to \mathbb{R}^m$, $g : \mathbb{R}^m \to \mathbb{R}^p$ and $h(x) = g(f(x))$:

$$\underbrace{Dh(x)}_{p\times n} = \underbrace{Dg(f(x))}_{p\times m}\ \underbrace{Df(x)}_{m\times n}.$$

Outer derivative first, evaluated at the inner value; then the inner derivative. With $m = p = 1$ this is $h'(x) = g'(f(x))f'(x)$.

WHY A PRODUCT OF MATRICES

A small step Δx moves f by about $Df\,\Delta x$; that moves g by about $Dg\,(Df\,\Delta x)$. Linear maps compose by multiplication, and the inner sizes (m) must match, like a relay where each runner hands over to the next.

EXAMPLE 2.4: AFFINE COMPOSITION -- USED CONSTANTLY

For $h(x) = g(Ax + b)$ with $g : \mathbb{R}^m \to \mathbb{R}$ and A of size $m \times n$:

$$Dh(x) = Dg(Ax + b)\,A, \qquad \nabla h(x) = A^T\,\nabla g(Ax + b).$$

(Transpose the first identity: $(Dg\,A)^T = A^T(Dg)^T = A^T\nabla g$.)

EXAMPLE · THE LEAST-SQUARES GRADIENT

$h(x) = \|Ax - b\|_2^2$ is $g(u) = u^Tu$ applied to $u = Ax - b$. By Example 2.3 with $A = I$: $\nabla g(u) = 2u$. So

$$\nabla h(x) = A^T \cdot 2(Ax - b) = 2A^T(Ax - b).$$

This is the starting point of Chapter 3 (least squares).

> 1.6 The Hessian

DEF · HESSIAN

For $f : \mathbb{R}^n \to \mathbb{R}$ (twice differentiable), the Hessian is the $n \times n$ matrix of second partial derivatives,

$$[\nabla^2 f(x)]_{ij} = \frac{\partial^2 f(x)}{\partial x_i\,\partial x_j}.$$

It is the Jacobian of the gradient. If the second derivatives are continuous, the order of differentiation does not matter, so the Hessian is symmetric.

EXAMPLE · A HESSIAN WITH NUMBERS

$f(x_1, x_2) = x_1^2 x_2 + x_2^3$. Gradient: $(2x_1x_2,\ x_1^2 + 3x_2^2)$. Differentiate each entry again:

$$\nabla^2 f(x) = \begin{pmatrix} 2x_2 & 2x_1 \\ 2x_1 & 6x_2 \end{pmatrix}.$$

The two off-diagonal entries agree ($\partial_2\partial_1 f = \partial_1\partial_2 f = 2x_1$), as promised. For $f = x^T Ax$ with symmetric A, $\nabla f = 2Ax$ is linear, so $\nabla^2 f = 2A$ (constant).

HESSIAN UNDER AN AFFINE SUBSTITUTION

$$g(x) = f(Ax + b) \quad \Longrightarrow \quad \nabla^2 g(x) = A^T\, \nabla^2 f(Ax + b)\, A.$$

Why: the gradient is $\nabla g(x) = A^T \nabla f(Ax + b)$. Its Jacobian, by the chain rule applied to $x \mapsto \nabla f(Ax + b)$, is $A^T\, \nabla^2 f(Ax + b)\, A$. For least squares: $\nabla^2 \|Ax - b\|^2 = A^T(2I)A = 2A^T A$.

> 1.7 Taylor expansions

SECOND-ORDER APPROXIMATION

$$f(y) \approx f(x) + \nabla f(x)^T (y - x) + \tfrac{1}{2}(y - x)^T \nabla^2 f(x)(y - x).$$

The gradient term is the slope; the Hessian term is the curvature. The $\frac{1}{2}$ is the same as in $f(x + t) \approx f(x) + tf'(x) + \frac{1}{2}t^2 f''(x)$.

EXAMPLE · FIRST VS. SECOND ORDER

$f = x_1^2 x_2 + x_2^3$ at $x = (1, 1)$: $f = 2$, $\nabla f = (2, 4)$, $\nabla^2 f = \begin{pmatrix} 2 & 2 \\ 2 & 6 \end{pmatrix}$. Step $d = (0.1, -0.1)$.

- First order: $2 + (2 \cdot 0.1 - 4 \cdot 0.1) = 1.8$.
- Curvature term: $\frac{1}{2} d^T \nabla^2 f d = \frac{1}{2}(2 \cdot 0.01 + 2 \cdot 2 \cdot (0.1)(-0.1) + 6 \cdot 0.01) = \frac{1}{2}(0.04) = 0.02$. Second order: 1.82.
- Exact: $f(1.1, 0.9) = 1.21 \cdot 0.9 + 0.729 = 1.818$.

Error $0.2 \to 0.002$: one more term made it a hundred times better.

EXACT REMAINDER FORM (USED IN EVERY PROOF BELOW)

There is a point $z = x + \alpha(y - x)$ for some $\alpha \in [0, 1]$ (on the segment between x and y) such that

$$f(y) = f(x) + \nabla f(x)^T (y - x) + \tfrac{1}{2}(y - x)^T \nabla^2 f(z)(y - x).$$

No "≈": the equation is exact, at the price of evaluating the Hessian at an unknown point z between x and y.

EXAMPLE · FINDING THE MYSTERY POINT

$f(x) = e^x$, $x = 0$, $y = 1$. The exact form says $e = 1 + 1 + \frac{1}{2}e^z$, so $e^z = 2(e-2) \approx 1.437$ and $z = \ln 1.437 \approx 0.36$, indeed in $[0, 1]$. We never need to know z; we only use facts that hold for *every* point of the segment (e.g. "$f'' \geq 0$ everywhere").

SECTION 02

Optimization

> 2.1 Vocabulary

DEF · AN OPTIMIZATION PROBLEM

$$\min_{x \in S} f(x)$$

- f is the objective function (the goal, the cost we want small).
- S is the feasible set (the constraints: the inputs we are allowed to use). A point $x \in S$ is feasible.
- A feasible x^* with the smallest objective is a minimizer; $f(x^*)$ is the optimal value.

Example: $\min_{x \in [0,3]} (x-1)^2$: objective $(x-1)^2$, feasible set $[0, 3]$, minimizer $x^* = 1$, optimal value 0. Maximizing f is the same as minimizing $-f$, so we only talk about minimizing. [BV §1.1, §4.1.1]

PITFALL · A MINIMIZER NEED NOT EXIST

- $\min_{x \in \mathbb{R}} x$: you can always go lower. No minimizer, the values are unbounded below.
- $\min_{x \in \mathbb{R}} e^x$: the values approach 0 but never reach it. The "best value" 0 is an infimum (greatest lower bound), not a minimum.
- $\min_{x \in (0,1]} x^2$: same problem, because the point 0 was cut out of the feasible set.

The rigorous fix is to write inf instead of min; the course avoids this formality, but keep in mind that "min" silently assumes a minimizer exists.

What counts as "solving"? In practice, finding a feasible x whose objective is within a prescribed tolerance (say 10^{-6}) of the optimum. **When is a problem easy?** The course measures this by the number of iterations (or "oracle calls": evaluations of f, ∇f, ...) an algorithm needs. "Just plot it" does not scale: in $n = 100$ dimensions, a grid with only 10 points per axis has 10^{100} points.

> 2.2 Four problems of this form

The lecture's motivating examples, all of the form $\min_{x \in S} f(x)$, all solved later in the course:

PROBLEM	FORMULATION	WHERE IN THE COURSE
Image reconstruction	$\min_x \|y - Hx\|^2$ s.t. $\|x\|_1 \le \alpha$	least squares (Ch. 3); sparsity (Ch. 5, 13)
Portfolio design	$\min_w w^T \Sigma w$ s.t. $w^T \mu = r,\ w^T \mathbf{1} = 1$	minimize risk at a target return
Support vector machine	$\min_{w,b} \|w\|^2$ s.t. $y_i(w^T x_i + b) \ge 1$	duality (Ch. 12)
Two-way partitioning	$\min_w w^T Q w$ s.t. $w_i \in \{\pm 1\}$	**hard**: bounds via relaxation (Ch. 12–13)

The last one looks just as simple, but its feasible set is 2^n isolated points: brute force takes exponential time. The difference between the first three and the fourth is, essentially, convexity.

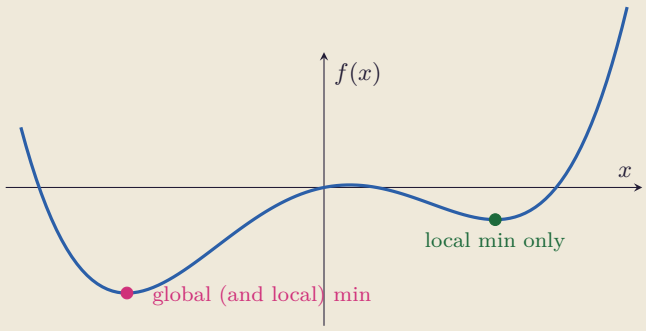
> 2.3 Global and local minima

DEF · DEFINITION 2.5: GLOBAL MINIMUM

$x^* \in S$ is a global minimum of f over S if it is no worse than *every* feasible point: $f(x^*) \le f(x)$ for all $x \in S$.

DEF · DEFINITION 2.6: LOCAL MINIMUM

$x^* \in S$ is a local minimum if it is no worse than its feasible *neighbours*: there is an $\epsilon > 0$ with $f(x^*) \le f(x)$ for all $x \in S$ with $\|x - x^*\| < \epsilon$.



$f(x) = x^4 - 3x^2 + x$. At $x \approx 1.13$ every *nearby* point is higher, so it is a local minimum, but $x \approx -1.30$ is lower still. A downhill-walking algorithm started at $x = 1.5$ gets stuck at the green point.

Every global minimum is a local one, not the other way round. Algorithms only ever "see" the neighbourhood of where they are, which is why we need conditions that are *local* (derivatives) and then a property (convexity) that turns local into global.

> 2.4 Necessary optimality conditions

THEOREM 2.7: NECESSARY CONDITIONS, SCALAR CASE

Let f be twice continuously differentiable and x^* a local minimum of f over an interval S. Then either

- x^* is an **endpoint** of S, or
- $f'(x^*) = 0$ **and** $f''(x^*) \geq 0$.

Proof, step by step. Assume x^* is a local minimum that is not an endpoint, so we can step a little both left and right.

1. **Local optimality, written down:** for all small $|\alpha|$, $f(x^* + \alpha) - f(x^*) \geq 0$.
2. **Divide by α.** For $\alpha > 0$ the quotient $\dfrac{f(x^* + \alpha) - f(x^*)}{\alpha} \geq 0$; for $\alpha < 0$ (dividing by a negative flips the sign) it is ≤ 0.
3. **Take the limit.** Both quotients tend to $f'(x^*)$. A limit of numbers ≥ 0 is ≥ 0, a limit of numbers ≤ 0 is ≤ 0. So $f'(x^*) = 0$.
4. **Exact Taylor.** $f(x^* + \alpha) - f(x^*) = \alpha f'(x^*) + \frac{\alpha^2}{2} f''(z)$ for some z between x^* and $x^* + \alpha$. The first term is 0 by step 3.
5. **Use optimality again:** $0 \leq f(x^* + \alpha) - f(x^*) = \frac{\alpha^2}{2} f''(z)$, so $f''(z) \geq 0$.
6. **Shrink $\alpha \to 0$.** Then $z \to x^*$, and by continuity of f'', $f''(x^*) \geq 0$. □

IN WORDS

At an interior bottom, the ground is flat ($f' = 0$; if it sloped, one side would be lower) and it curves up, or at least not down ($f'' \geq 0$). At an endpoint you are not allowed to step outside, so the slope can be anything.

EXAMPLE · THE ENDPOINT CLAUSE MATTERS

$\min_{x \in [0,1]} x$: the minimizer is $x^* = 0$, but $f'(0) = 1 \neq 0$. No contradiction: 0 is an endpoint, and stepping left (where f is lower) is infeasible.

PITFALL · NECESSARY, NOT SUFFICIENT

$f(x) = x^3$ at $x = 0$: $f'(0) = 0$ and $f''(0) = 0 \geq 0$, so both conditions hold, yet 0 is not a local minimum ($f(-0.1) = -0.001 < 0$). The theorem only *rules out* candidates; points that pass are just candidates.

Straightforward variations (from the lecture):

- **Maximization:** reverse the inequalities: $f'(x^*) = 0$, $f''(x^*) \leq 0$.
- **Multivariate:** $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ (PSD, Week 01).

Why a matrix condition? Run the scalar proof along *every* line through x^*. Fix a direction d and look at $\alpha \mapsto f(x^* + \alpha d)$:

$$f(x^* + \alpha d) - f(x^*) = \alpha \nabla f(x^*)^T d + \tfrac{\alpha^2}{2} d^T \nabla^2 f(z)\, d.$$

The scalar argument gives $\nabla f(x^*)^T d = 0$ for every d (so $\nabla f(x^*) = 0$) and $d^T \nabla^2 f(x^*) d \geq 0$ for every d, which is exactly the definition of $\nabla^2 f(x^*) \succeq 0$.

EXAMPLE · A SADDLE POINT

$f(x_1, x_2) = x_1^2 - x_2^2$: $\nabla f = (2x_1, -2x_2) = 0$ only at the origin, and $\nabla^2 f = \mathrm{Diag}(2, -2)$, which is not PSD ($d = (0, 1)$ gives -2). So the origin is **not** a local minimum: along the x_2-axis, $f = -x_2^2$ goes down. This shape (up in one direction, down in another) is called a saddle point.

PITFALL · THE LECTURE'S "STRICT" VARIATION, READ CAREFULLY

The notes say that for strict local minima one "replaces the inequalities by strict ones". That is right for the **sufficient** direction: if $f'(x^*) = 0$ and $f''(x^*) > 0$, then x^* is a strict local minimum. It is **not** a necessary condition: $f(x) = x^4$ has a strict local (even global) minimum at 0, but $f''(0) = 0$, not > 0.

SECTION 03

Convexity

The conditions so far are *local*: they cannot tell the green point of the figure from the pink one. Convexity is the property that makes local information global. **This week everything is in one dimension**; convexity in $\mathbb{R}^n$ is Chapters 7–8.

> 3.1 Convex sets

DEF · DEFINITION 2.8: CONVEX SET

A set S is convex if for all $x, y \in S$ and all $\lambda \in [0, 1]$, the point $\lambda x + (1 - \lambda) y$ is also in S.

READING THE FORMULA

$\lambda x + (1 - \lambda) y$ for λ from 0 to 1 walks along the straight segment from y (at $\lambda = 0$) to x (at $\lambda = 1$); $\lambda = \frac{1}{2}$ is the midpoint. So: a set is convex if, with any two of its points, it contains the whole segment between them. No holes, no dents. Example: $x = 4$, $y = 0$, $\lambda = \frac{1}{4}$ gives $\frac{1}{4} \cdot 4 + \frac{3}{4} \cdot 0 = 1$, a quarter of the way from 0 to 4.

In one dimension the segment between x and y is just the interval between them, so **the convex subsets of $\mathbb{R}$ are exactly the intervals**: $\{x \geq a\}$, $\{a \leq x \leq b\}$, $\{x \leq b\}$, open or closed ends, and $\mathbb{R}$ itself. [BV §2.1.4]

EXAMPLE · CONVEX OR NOT?

$[0, 5]$, $(-\infty, 2)$, $\{3\}$, $\mathbb{R}$: convex. $[0, 1] \cup [2, 3]$: **not** convex, since 1 and 2 are in it but their midpoint 1.5 is not.

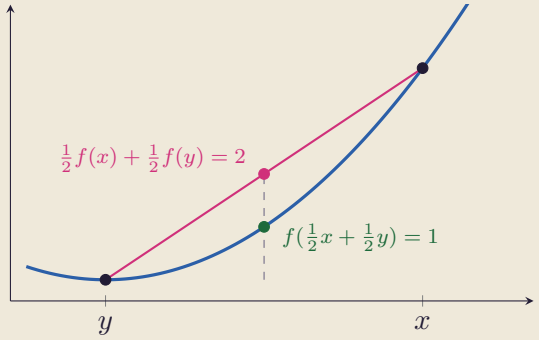
> 3.2 Convex functions

DEF · DEFINITION 2.9: CONVEX FUNCTION

f is convex over an interval I if for all $x, y \in I$ and $\lambda \in [0,1]$:

$$f(\lambda x + (1-\lambda)y) \;\le\; \lambda f(x) + (1-\lambda) f(y).$$

f is concave if $-f$ is convex (reverse the inequality). The left side needs $\lambda x + (1-\lambda)y \in I$: that is why the domain must be an interval (a convex set). [BV §3.1.1]



$f(x) = x^2$, $y = 0$, $x = 2$, $\lambda = \frac{1}{2}$. The left side of the definition is the graph at the midpoint (green, 1); the right side is the chord at the midpoint (pink, 2). Convex: every chord lies above the graph.

A PASTRY PICTURE

Lay a strip of Apfelstrudel dough over a bowl and pull it tight between any two points of the rim: the tight strip (the chord) never dips below the bowl's surface (the graph). A bowl is convex; a bumpy surface (like the figure in §2.3) is not, because a chord can pass *under* a bump.

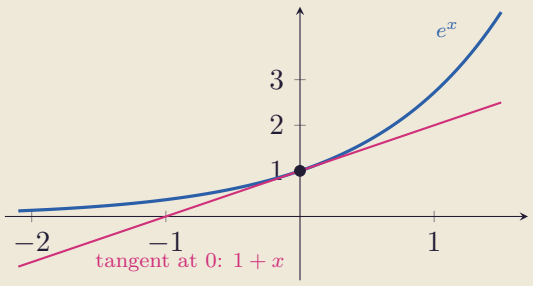
> 3.3 First-order condition: above every tangent

THEOREM 2.10: FIRST-ORDER CONDITION

A continuously differentiable f is convex over I **if and only if**

$$f(x) \;\ge\; f(y) + (x-y) f'(y) \qquad \text{for all } x, y \in I.$$

The right side is the tangent line at y, evaluated at x. So: a convex function lies above all of its tangent lines. [BV §3.1.3]



$e^x \geq 1 + x$ for every x: the first-order condition at $y = 0$ ($f(0) = 1$, $f'(0) = 1$).

Proof, step by step.

(⇒) Convex implies above tangents.

1. Fix $x \neq y$ in I (if $x = y$ there is nothing to prove). Write the definition with the point $y + \lambda(x - y) = \lambda x + (1 - \lambda)y$: $f(y + \lambda(x - y)) \leq \lambda f(x) + (1 - \lambda)f(y)$, for $\lambda \in (0, 1]$.
2. Subtract $f(y)$ and divide by $\lambda > 0$: $\dfrac{f(y + \lambda(x - y)) - f(y)}{\lambda} \leq f(x) - f(y)$.
3. Write the left side as $\dfrac{f(y + \lambda(x - y)) - f(y)}{\lambda(x - y)} \cdot (x - y)$. The fraction is a difference quotient with step $\lambda(x - y)$.
4. Let $\lambda \to 0$: the fraction tends to $f'(y)$. So $f'(y)(x - y) \leq f(x) - f(y)$, which rearranges to the claim.

(⇐) Above tangents implies convex.

1. Fix x, y, $\lambda \in [0, 1]$, and the point between them $z = \lambda x + (1 - \lambda)y$.
2. Use the tangent at z twice: $f(x) \geq f(z) + (x - z)f'(z)$ and $f(y) \geq f(z) + (y - z)f'(z)$.
3. Multiply the first by λ, the second by $1 - \lambda$, and add: $\lambda f(x) + (1 - \lambda)f(y) \geq f(z) + \big[\lambda x + (1 - \lambda)y - z\big]f'(z)$.
4. The bracket is $z - z = 0$. What remains, $\lambda f(x) + (1 - \lambda)f(y) \geq f(z)$, is the definition. □

WHY THIS IS THE MOST USEFUL FORM

The tangent line is built from information at *one* point $(f(y), f'(y))$, yet the inequality holds at *every* x. For a convex function, local information gives a global lower bound. In particular, if $f'(y) = 0$, then $f(x) \geq f(y)$ for every x: a flat point of a convex function is a global minimum (Problem P15).

> 3.4 Second-order condition: curving up everywhere

THEOREM 2.11: SECOND-ORDER CONDITION

A twice continuously differentiable f is convex over I **if and only if** $f''(x) \geq 0$ for all $x \in I$. [BV §3.1.4]

Proof, step by step.

1. *Sufficiency.* Exact Taylor: $f(y) - f(x) = (y - x)f'(x) + \frac{(y-x)^2}{2}f''(z)$ for some z between x and y.
2. If $f'' \geq 0$ everywhere, the last term is ≥ 0, so $f(y) \geq f(x) + (y - x)f'(x)$: the first-order condition. By Theorem 2.10, f is convex.
3. *Necessity.* Suppose f is convex but $f''(x) < 0$ at some x. Since f'' is continuous, $f'' < 0$ on a small interval around x; pick $y \neq x$ in it.
4. Then every z between x and y has $f''(z) < 0$, so the Taylor formula gives $f(y) < f(x) + (y - x)f'(x)$: the graph is *below* a tangent. This contradicts Theorem 2.10. □

This is the *practical* test: one derivative computation, one sign check.

FUNCTION	DOMAIN	$f''(x)$	VERDICT
e^x	$\mathbb{R}$	$e^x > 0$	convex
x^a	$\mathbb{R}_{++} = (0, \infty)$	$a(a-1)x^{a-2}$	convex: $a \geq 1$ or $a \leq 0$ concave: $0 \leq a \leq 1$
$-\log x$	$\mathbb{R}_{++}$	$1/x^2 > 0$	convex (so $\log x$ is concave)
$x \log x$ (negative entropy)	$\mathbb{R}_{++}$	$1/x > 0$	convex

Reading the power row. $x^{a-2} > 0$ on $\mathbb{R}_{++}$, so the sign of f'' is the sign of $a(a-1)$: positive outside $[0, 1]$, negative inside. Examples: x^2 ($a = 2$, convex), $1/x$ ($a = -1$, $f'' = 2/x^3 > 0$, convex), $\sqrt{x}$ ($a = \frac{1}{2}$, $f'' = -\frac{1}{4}x^{-3/2} < 0$, concave), x ($a = 1$, both). [BV §3.1.5]

PITFALL · $f'' \geq 0$, NOT $f'' > 0$

$f(x) = x^4$ is convex on $\mathbb{R}$ even though $f''(0) = 0$. The condition is "≥ 0 everywhere". (Conversely, one point with $f'' < 0$ is enough to break convexity.)

PITFALL · THE TESTS NEED DERIVATIVES; THE DEFINITION DOES NOT

$f(x) = |x|$ is convex (every chord lies on or above the V), but $f'(0)$ does not exist, so Theorems 2.10–2.11 cannot be applied at 0. Use the definition, or the triangle inequality: $|\lambda x + (1-\lambda)y| \leq \lambda|x| + (1-\lambda)|y|$.

PITFALL · THE DOMAIN IS PART OF THE ANSWER

x^3 has $f'' = 6x$: convex on $[0, \infty)$, **not** convex on $\mathbb{R}$. Always state the interval. Similarly the power table is about $\mathbb{R}_{++}$ only: x^{-1} is not convex on $\mathbb{R} \setminus \{0\}$ (that set is not even an interval).

> 3.5 The payoff: local = global

THEOREM 2.12: LOCAL = GLOBAL

If f is convex over an interval I, then every local minimum of f over I is a global minimum. [BV §4.2.2]

Proof, step by step (by contradiction).

1. Suppose x is a local minimum but not global: some $y \in I$ has $f(y) < f(x)$.
2. Walk from x toward y: the points $\lambda x + (1-\lambda)y$ with λ close to 1 are close to x (and in I, because I is an interval).
3. Convexity: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$.
4. Because $f(y) < f(x)$ and $1 - \lambda > 0$, the right side is *strictly* less than $\lambda f(x) + (1-\lambda)f(x) = f(x)$.
5. So as $\lambda \to 1$ we find points arbitrarily close to x with $f < f(x)$. That contradicts x being a local minimum. □

WHY THE PROOF WORKS

The chord from x down to the lower point y starts at height $f(x)$ and goes strictly down. A convex function stays *below* its chords, so it must also go down right next to x. A convex function cannot hide a lower valley behind a hill: there are no hills.

EXAMPLE · SEEING IT ON THE FIGURE OF §2.3

$f(x) = x^4 - 3x^2 + x$ has a local minimum that is not global, so by Theorem 2.12 it cannot be convex. Check: $f''(x) = 12x^2 - 6$, and $f''(0) = -6 < 0$. ✓

Remark (from the lecture): convexity is sufficient, not necessary. Some non-convex functions also have local = global, for example $f(x) = \sqrt{|x|}$: its only local minimum is $x = 0$, which is global, but $f''(x) = -\frac{1}{4}|x|^{-3/2} < 0$ for $x \neq 0$ (concave on each side). Such functions are *(strictly) quasi-convex*. Convexity is still the property we build on: it is the easiest to verify, the best understood (no saddle points, preserved under sums), and it generalizes cleanly to $\mathbb{R}^n$.

WEEK 02

Practice Problems and Summary

SECTION 04

Practice problems with full solutions

Try each problem before reading its solution. ● ○ ○ easy, ● ● ○ medium, ● ● ● hard.

> 4.1 Derivatives

EXAMPLE · P1 ● ○ ○: JACOBIAN AND LINEAR APPROXIMATION

$f(x_1, x_2) = (x_1 x_2,\ x_1^2 - x_2,\ 3x_2)$. (a) What size is Df? (b) Compute $Df(2, 1)$. (c) Approximate $f(2.1,\ 0.9)$ and compare with the exact value.

Solution. (a) 3 outputs, 2 inputs: 3×2. (b) Row by row: $\nabla f_1 = (x_2, x_1)$, $\nabla f_2 = (2x_1, -1)$, $\nabla f_3 = (0, 3)$. At $(2, 1)$: $Df = \begin{pmatrix} 1 & 2 \\ 4 & -1 \\ 0 & 3 \end{pmatrix}$. (c) $f(2, 1) = (2, 3, 3)$, step $(0.1, -0.1)$: $Df \cdot (0.1, -0.1) = (0.1 - 0.2,\ 0.4 + 0.1,\ -0.3) = (-0.1, 0.5, -0.3)$, so $f \approx (1.9, 3.5, 2.7)$. Exact: $(2.1 \cdot 0.9,\ 4.41 - 0.9,\ 2.7) = (1.89,\ 3.51,\ 2.7)$. The third output is linear, so it is exact.

EXAMPLE · P2 ● ○ ○: GRADIENT OF A NON-SYMMETRIC QUADRATIC FORM

$A = \begin{pmatrix} 3 & 4 \\ 0 & 1 \end{pmatrix}$, $f(x) = x^T A x$. Compute $\nabla f(1, -1)$ in two ways, and find the symmetric matrix S with $x^T S x = x^T A x$.

Solution. Expanded: $f = 3x_1^2 + 4x_1 x_2 + x_2^2$, so $\nabla f = (6x_1 + 4x_2,\ 4x_1 + 2x_2)$, at $(1, -1)$: $(2, 2)$. Formula: $A + A^T = \begin{pmatrix} 6 & 4 \\ 4 & 2 \end{pmatrix}$, times $(1, -1)$: $(2, 2)$ ✓. $S = \frac{1}{2}(A + A^T) = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$, and then $\nabla f = 2Sx$.

EXAMPLE · P3 ● ● ○: LEAST SQUARES BY HAND

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, $b = (1, 1, 0)$, $h(x) = \|Ax - b\|^2$. Compute ∇h and $\nabla^2 h$, and find the stationary point. Is it a minimum?

Solution. By the affine chain rules: $\nabla h = 2A^T(Ax - b)$, $\nabla^2 h = 2A^T A$. $A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $A^T b = (1, 1)$. Setting $\nabla h = 0$: $A^T A x = A^T b$, i.e. $2x_1 + x_2 = 1$, $x_1 + 2x_2 = 1$, so $x^* = (\frac{1}{3}, \frac{1}{3})$. The Hessian $2A^T A$ has eigenvalues $6, 2 > 0$ (Week 01 example), so it is PD: the necessary

conditions hold, and (Chapters 7–8) this is the global minimum. Value: $Ax^* - b = (-\frac{2}{3}, -\frac{2}{3}, \frac{2}{3})$, $h(x^*) = \frac{4}{3}$.

EXAMPLE · P4 ●●○: AN EXPONENTIAL OF A LINEAR FUNCTION

$h(x) = e^{a^T x}$ with $a \in \mathbb{R}^n$. Find ∇h and $\nabla^2 h$. Show $\nabla^2 h \succeq 0$. Evaluate both at $x = 0$ for $a = (1, 2)$.

Solution. $h = g(a^T x)$ with $g(t) = e^t$, and $a^T x$ is affine with "A" $= a^T$ $(1 \times n)$. So $\nabla h = a\, g'(a^T x) = e^{a^T x} a$ and $\nabla^2 h = a\, g''(a^T x)\, a^T = e^{a^T x} a a^T$. For any v: $v^T (e^{a^T x} a a^T) v = e^{a^T x} (a^T v)^2 \geq 0$. At $x = 0$: $\nabla h = (1, 2)$, $\nabla^2 h = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ (PSD, singular: $\det = 0$).

EXAMPLE · P5 ●●○: SECOND-ORDER TAYLOR

$f(x_1, x_2) = e^{x_1} \cos x_2$. Write the second-order expansion around $(0, 0)$ and use it to approximate $f(0.1, 0.2)$.

Solution. $f(0) = 1$. $\nabla f = (e^{x_1} \cos x_2,\ -e^{x_1} \sin x_2) = (1, 0)$ at 0.

$$\nabla^2 f = \begin{pmatrix} e^{x_1} \cos x_2 & -e^{x_1} \sin x_2 \\ -e^{x_1} \sin x_2 & -e^{x_1} \cos x_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ at } 0.$$

So $f \approx 1 + x_1 + \frac{1}{2}(x_1^2 - x_2^2)$. At $(0.1, 0.2)$: $1 + 0.1 + \frac{1}{2}(0.01 - 0.04) = 1.085$. Exact: $e^{0.1} \cos 0.2 \approx 1.10517 \cdot 0.98007 \approx 1.0832$.

> 4.2 Optimality

EXAMPLE · P6 ●○○: DOES A MINIMIZER EXIST?

(a) $\min_{x \geq 0} e^{-x}$; (b) $\min_{x \in [-1,1]} x^2$; (c) $\min_{x \in \mathbb{R}} x^3$; (d) $\min_{x > 0} (x + 1/x)$.

Solution. (a) No: $e^{-x} \to 0$ but never reaches 0 (infimum 0). (b) Yes, $x^* = 0$. (c) No: unbounded below. (d) Yes: $f' = 1 - 1/x^2 = 0$ at $x = 1$; $f'' = 2/x^3 > 0$ on $x > 0$, so f is convex there and $x^* = 1$ is global (Theorem 2.12), $f(1) = 2$.

EXAMPLE · P7 ●●○: INTERIOR POINTS AND ENDPOINTS

Find all local and global minima and maxima of $f(x) = x^3 - 3x$ on $[-3, 3]$.

Solution.

1. Interior candidates: $f' = 3x^2 - 3 = 0$ at $x = \pm 1$. $f'' = 6x$: $f''(1) = 6 > 0$, local min, $f(1) = -2$; $f''(-1) = -6 < 0$, local max, $f(-1) = 2$.
2. Endpoints: $f(-3) = -27 + 9 = -18$, $f(3) = 27 - 9 = 18$.
3. Compare: global min at the endpoint $x = -3$ (value -18), global max at $x = 3$ (18).

Note $f'(-3) = 24 \neq 0$ at the global minimum: allowed, by the endpoint clause of Theorem 2.7. And $x = 1$ is local but not global, so f is not convex on $[-3, 3]$ (indeed $f'' < 0$ for $x < 0$).

EXAMPLE · P8 ●●○: CLASSIFY THE STATIONARY POINTS

(a) $f = x_1^2 + 4x_1x_2 + x_2^2$. (b) $f = x_1^2 + x_1x_2 + x_2^2 - 3x_1$.

Solution. (a) $\nabla f = (2x_1+4x_2,\ 4x_1+2x_2) = 0$ only at 0. $\nabla^2 f = \begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$, eigenvalues 6 and -2: not PSD, so 0 is a saddle, not a minimum. Along $d = (1,-1)$: $f(t,-t) = t^2-4t^2+t^2 = -2t^2 < 0$. (b) $\nabla f = (2x_1 + x_2 - 3,\ x_1 + 2x_2) = 0$: from the second, $x_1 = -2x_2$; then $-3x_2 = 3$, so $x^* = (2,-1)$. $\nabla^2 f = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \succ 0$. The necessary conditions hold; $f(2,-1) = 4-2+1-6 = -3$.

EXAMPLE · P9 ●○○: NECESSARY IS NOT SUFFICIENT

For each f, check whether $x = 0$ satisfies $f'(0) = 0$, $f''(0) \geq 0$, and whether it is a local minimum: (a) x^4; (b) $-x^4$; (c) x^3; (d) x^2.

Solution. All four have $f'(0) = 0$. $f''(0)$: (a) 0, (b) 0, (c) 0, (d) 2. So all four *pass* the necessary test. Local min? (a) yes, (b) no (it is a max), (c) no, (d) yes. The test cannot separate (a) from (b) and (c).

> 4.3 Convexity

EXAMPLE · P10 ●○○: CONVEX SETS IN $\mathbb{R}$

Which are convex: (a) $[2,7]$; (b) $\{x : x^2 \leq 4\}$; (c) $\{x : x^2 \geq 4\}$; (d) the integers $\mathbb{Z}$; (e) $\{x : |x-1| < 3\}$?

Solution. (a) yes. (b) $= [-2,2]$, yes. (c) $= (-\infty,-2] \cup [2,\infty)$, no: -2 and 2 are in it, their midpoint 0 is not. (d) no: $0, 1 \in \mathbb{Z}$ but $\frac{1}{2} \notin \mathbb{Z}$. (e) $= (-2,4)$, yes.

EXAMPLE · P11 ●●○: CONVEXITY OF x^2 FROM THE DEFINITION

Prove that $f(x) = x^2$ is convex on $\mathbb{R}$ using only Definition 2.9 (no derivatives).

Solution. Compute right side minus left side:

$$\lambda x^2+(1-\lambda)y^2-(\lambda x+(1-\lambda)y)^2 = \lambda(1-\lambda)x^2+\lambda(1-\lambda)y^2-2\lambda(1-\lambda)xy = \lambda(1-\lambda)(x-y)^2 \geq 0,$$

using $\lambda - \lambda^2 = \lambda(1-\lambda)$ and $(1-\lambda) - (1-\lambda)^2 = \lambda(1-\lambda)$. Both factors are ≥ 0 for $\lambda \in [0,1]$. Check with the figure: $\lambda = \frac{1}{2}$, $x = 2$, $y = 0$: $\frac{1}{4} \cdot 4 = 1 = 2 - 1$. ✓

EXAMPLE · P12 ●○○: THE SECOND-ORDER TEST

Convex, concave, or neither, on the given domain? (a) $x^2 + 1/x$ on $\mathbb{R}_{++}$; (b) $\sqrt{x}$ on $\mathbb{R}_{++}$; (c) e^{x^2} on $\mathbb{R}$; (d) x^3 on $\mathbb{R}$; (e) $\log x - x^2$ on $\mathbb{R}_{++}$; (f) $\cos x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

Solution. (a) $f'' = 2 + 2/x^3 > 0$: convex. (b) $f'' = -\frac{1}{4}x^{-3/2} < 0$: concave. (c) $f' = 2xe^{x^2}$, $f'' = (2+4x^2)e^{x^2} > 0$: convex. (d) $f'' = 6x$ changes sign: neither. (e) $f'' = -1/x^2 - 2 < 0$: concave. (f) $f'' = -\cos x \leq 0$ there: concave.

EXAMPLE · P13 ●●○: AN INEQUALITY FROM A TANGENT LINE

Use the first-order condition for $f(x) = -\log x$ at $y = 1$ to prove $\log x \le x - 1$ for all $x > 0$. When is it an equality?

Solution. f is convex on $\mathbb{R}_{++}$ (table), $f(1) = 0$, $f'(1) = -1/1 = -1$. Theorem 2.10: $-\log x \ge 0 + (x-1)(-1) = 1 - x$, i.e. $\log x \le x - 1$. Equality at $x = 1$ (touching point). Numbers: $x = 2$: $\log 2 \approx 0.693 \le 1$; $x = 0.5$: $-0.693 \le -0.5$. ✓

EXAMPLE · P14 ●●○: TWO VALLEYS

$f(x) = x^4 - 2x^2$. Find all local minima and maxima. Is f convex on $\mathbb{R}$? On $[1, \infty)$? Draw the chord that breaks convexity.

Solution. $f' = 4x(x^2 - 1) = 0$ at $0, \pm 1$. $f'' = 12x^2 - 4$: $f''(0) = -4$, local max ($f = 0$); $f''(\pm 1) = 8$, local minima ($f = -1$, both global). Not convex on $\mathbb{R}$: $f''(0) < 0$. Chord from -1 to 1: at the midpoint, chord height $\frac{1}{2}(-1) + \frac{1}{2}(-1) = -1$, but $f(0) = 0 > -1$: the graph is above the chord. On $[1, \infty)$: $f'' \ge 12 - 4 = 8 > 0$, convex. (In fact $f'' \ge 0$ exactly when $|x| \ge 1/\sqrt{3}$.)

EXAMPLE · P15 ●●●: A FLAT POINT OF A CONVEX FUNCTION IS A GLOBAL MINIMUM

Let f be differentiable and convex on an interval I, and $f'(x^*) = 0$ for an interior x^*. Prove x^* is a global minimum, without using Theorem 2.12.

Solution. Theorem 2.10 with $y = x^*$: $f(x) \ge f(x^*) + (x - x^*) \cdot 0 = f(x^*)$ for all $x \in I$. That is the definition of a global minimum. *So for convex f the necessary condition $f' = 0$ is also sufficient*, in contrast with P9 (x^3, $-x^4$ are not convex).

EXAMPLE · P16 ●●●: SUMS AND SCALINGS PRESERVE CONVEXITY

(a) If f, g are convex on I and $\alpha, \beta \ge 0$, show $\alpha f + \beta g$ is convex using the definition. (b) Use (a) and the table to show $h(x) = 3e^x - 2\log x + x \log x$ is convex on $\mathbb{R}_{++}$, and check with h''. (c) Is $f - g$ convex in general?

Solution. (a) Write $m = \lambda x + (1-\lambda)y$. Then $\alpha f(m) + \beta g(m) \le \alpha[\lambda f(x) + (1-\lambda) f(y)] + \beta[\lambda g(x) + (1-\lambda) g(y)] = \lambda(\alpha f + \beta g)(x) + (1-\lambda)(\alpha f + \beta g)(y)$. The inequality needs $\alpha, \beta \ge 0$ (multiplying by a negative number would flip it). (b) $3e^x$, $2(-\log x)$ and $x \log x$ are nonnegative multiples of convex functions. Check: $h'' = 3e^x + 2/x^2 + 1/x > 0$. ✓(c) No: $f = x^2$, $g = 2x^2$ are convex, but $f - g = -x^2$ is concave.

SECTION 05

Cheat sheet

WEEK 02 ON ONE PAGE

- **Jacobian** $Df(x)$: $m \times n$, $[Df]_{ij} = \partial f_i/\partial x_j$. $f(z) \approx f(x) + Df(x)(z-x)$. $D(Ax) = A$.
- **Gradient** ($m = 1$): $\nabla f = (Df)^T$, a column. $f(z) \approx f(x) + \nabla f(x)^T(z-x)$; points uphill. $\nabla(a^Tx) = a$, $\nabla(x^TAx) = (A + A^T)x$ ($= 2Ax$ if symmetric).
- **Chain rule**: $D[g(f(x))] = Dg(f(x))\,Df(x)$. Affine: $\nabla g(Ax+b) = A^T\nabla g(\cdot)$, $\nabla^2 = A^T\nabla^2 g(\cdot)A$. Least squares: $\nabla\|Ax-b\|^2 = 2A^T(Ax-b)$, $\nabla^2 = 2A^TA$.
- **Hessian**: $[\nabla^2 f]_{ij} = \partial^2 f/\partial x_i \partial x_j$, symmetric. Taylor: $f(y) \approx f(x) + \nabla f^T(y-x) + \frac{1}{2}(y-x)^T\nabla^2 f(x)(y-x)$; exact with $\nabla^2 f(z)$, z on the segment.
- **Problem** $\min_{x\in S} f$: objective, feasible set. A minimizer may not exist ($\min x$, $\min e^x$). Global: $f(x^*) \le f(x)\ \forall x \in S$. Local: same, only for $\|x - x^*\| < \epsilon$.
- **Necessary conditions** (local min, interior): $f' = 0$, $f'' \ge 0$; in $\mathbb{R}^n$: $\nabla f = 0$, $\nabla^2 f \succeq 0$. At an endpoint anything goes. Not sufficient (x^3). $f' = 0$, $f'' > 0$ is sufficient for a strict local min.
- **Convex set**: contains every segment; in $\mathbb{R}$, exactly the intervals.
- **Convex function**: $f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda)f(y)$ (chord above graph). Concave: $-f$ convex.
- **Tests** (1-D): $f(x) \ge f(y) + (x-y)f'(y)$ (above tangents) $\iff$ convex $\iff$ $f'' \ge 0$ on I.
- **Examples** on $\mathbb{R}_{++}$: e^x (on $\mathbb{R}$), x^a ($a \ge 1$ or $a \le 0$), $-\log x$, $x\log x$ convex; x^a ($0 \le a \le 1$), $\log x$ concave.
- **Local = global** for convex f; and for convex differentiable f, $f'(x^*) = 0 \Rightarrow$ global min.